

Protocol A-SYNC: From Divergence to Convergence on Deep Models

值域散失所创造pU

Algorithm Variant Report / Bilingual (EN + ZH)

算法变体报告 / 中英双语

Protocol A (ALS -> SGD -> Perturb) diverged on all models with 28+ layers.
A-SYNC replaces direct weight writes with gradient injection.
Qwen2.5-7B (28L): PPL 58.8 -> 7.6 in 48 cycles. Monotonic convergence.

Protocol AALS->SGD->Perturb 28+
A-SYNC
Qwen2.5-7B (28L)PPL 58.8 -> 7.648

1. Motivation: Why Original Protocol A Fails

Protocol A interleaves three phases: ALS (exact block-wise least squares on lm_head), SGD (stochastic gradient descent on all parameters), and Perturb (random noise injection). On models with 12-24 transformer layers, this converges reliably. On models with 28+ layers, every attempt diverges within 2-3 cycles.

GD

值域散失所

Protocol A 交替执行三个阶段：ALS（对 lm_head 进行精确逐块最小二乘求解）、SGD（对所有参数进行随机梯度下降）和 Perturb（随机噪声注入）。在 12-24 层 Transformer 模型上，该过程可靠收敛。在 28 层及以上模型上，所有尝试均在 2-3 个周期内发散。

Root Cause: Residual Amplification

ALS m
blocks
27 res
alpha

ALS
块。每
阶段每

Why Protocol A Fails on Deep Models

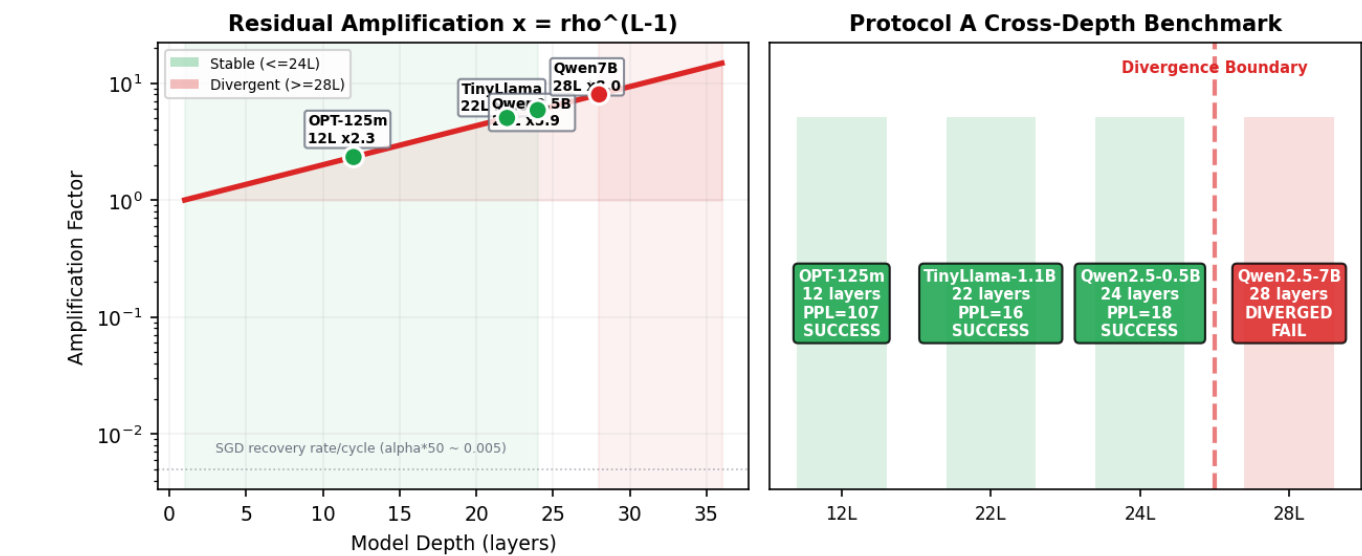


Figure 1: Left — residual amplification $\rho^{(L-1)}$ vs model depth with experimental data. Right — Protocol A cross-depth benchmark.

Model	Layers	Amplification	Protocol A PPL	Status
OPT-125m	12	2.3x	106.9	SUCCESS
TinyLlama-1.1B	22	5.0x	15.5	SUCCESS
Qwen2.5-0.5B	24	5.9x	18.0	SUCCESS (unstable)
Qwen2.5-7B	28	8.0x	DIVERGED	11/11 FAIL

Transformer Architecture: Residual Amplification Path in Protocol A

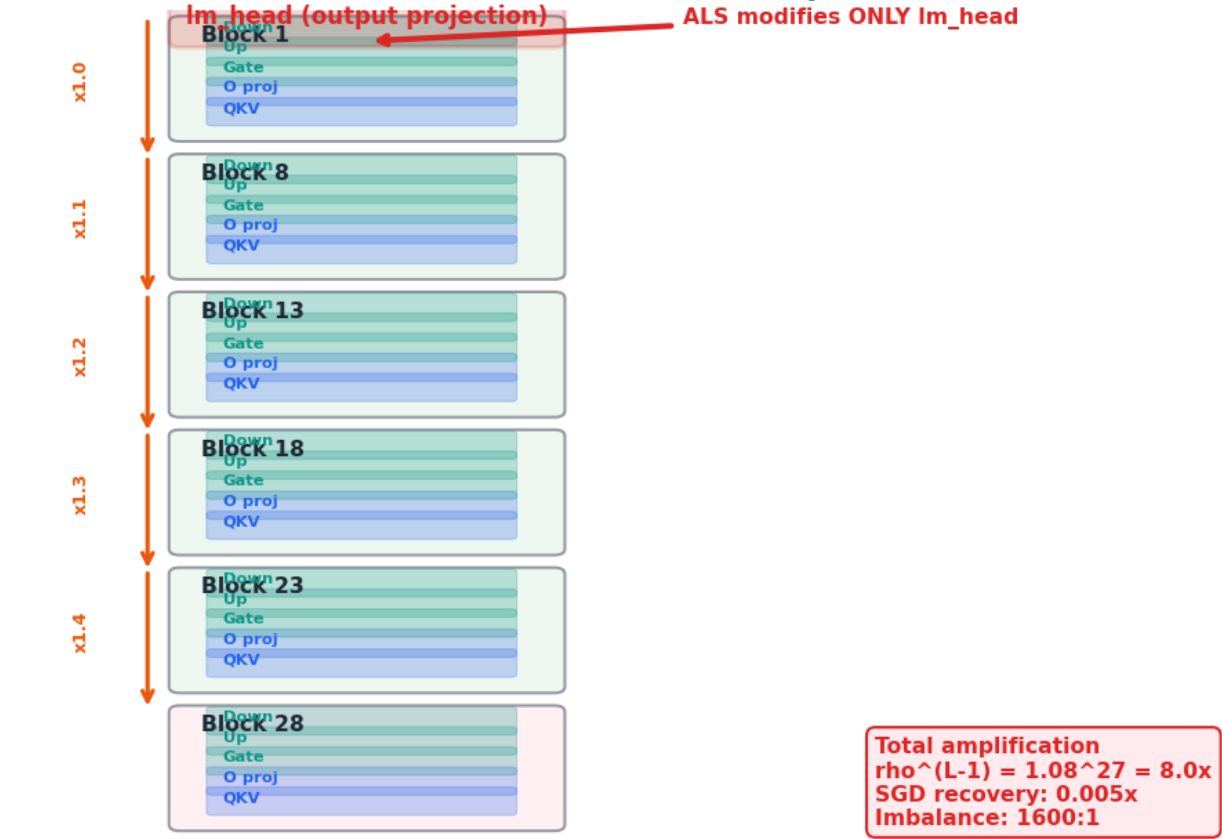


Figure 2: Transformer architecture — residual amplification path in Protocol A. ALS modifies only Im_head, perturbation propagates through L-1 frozen blocks.

The problem is structural, not parametric. ALS's Im_head-only design creates a 1600:1 amplification-to-recovery asymmetry

2. A-SYNC Algorithm: Gradient Injection

Core Innovation

1. 创新点

Protocol Pseudocode / :

1. ALS solve on lm_head -> get W_new (label-based exact least squares)
2. Compute delta = W_new - W_old (CPU offload to save GPU memory)
3. Revert lm_head to W_old
4. SGD for 50 steps: each step add sync_strength * delta to lm_head gradient
5. (Perturbation: REMOVED – causes oscillations)
6. Repeat from step 1

Protocol A-SYNC

1. ALS lm_head -> W_new
2. delta = W_new - W_oldCPU GPU
3. lm_head W_old
4. SGD 50 sync_strength * delta lm_head
5. Perturb —
6. 1

Final Configuration / :

sync_strength: 0.05 (CONSTANT, no decay)
learning_rate: 2e-4 (CONSTANT)
momentum: 0.0, weight_decay: 0.01
cycles: 24-48 (converges at ~44)
ALS: block_size=512, reg_lambda=1e-3, step_size=0.01
Perturbation: DISABLED

sync_strength: 0.05
learning_rate: 2e-4
momentum: 0.0, weight_decay: 0.01
cycles: 24-48 44
ALS: block_size=512, reg_lambda=1e-3, step_size=0.01
Perturb

A-SYNC gradient injection: ALS computes WHERE to go (direction). SGD handles HOW to get there (optimization). The two sig

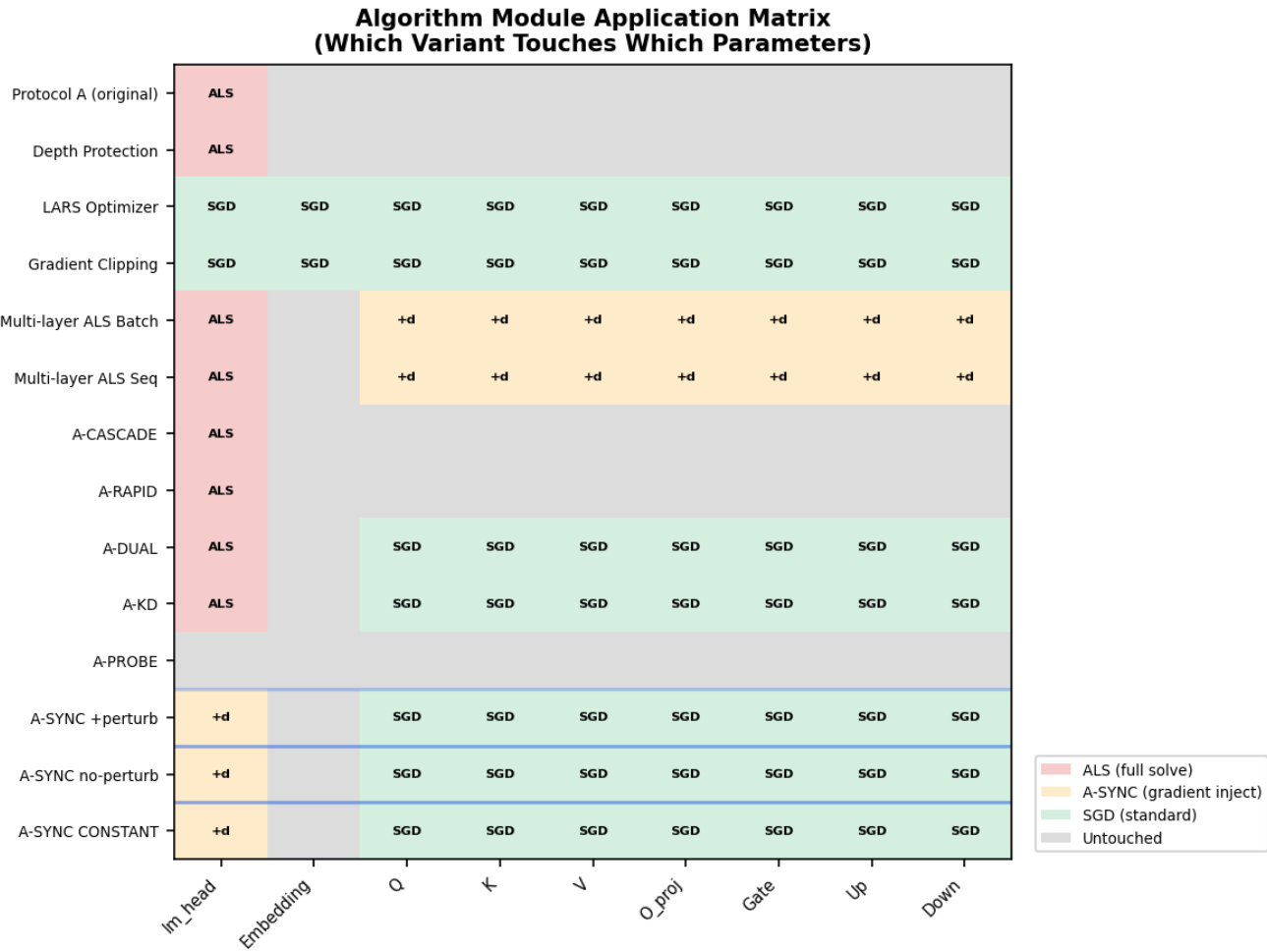


Figure 3: Algorithm Module Application Matrix — which variant touches which parameters. A-SYNC (bottom rows) only gradient-injects Im_head.

3. All Fix Attempts Explained

LD

Parameter Tuning (alpha, ALS:SGD ratio) [Tuning] — FAILED

将 ALS 学习率调低，将 ALS:SGD 比例调低

Reduced
alpha=
ALS:SGD
ratio
将 ALS
学习率
调低，
将 ALS:
SGD 比
例调低。

Depth-Boundary Protection [Protection] — PARTIAL

将深度边界保护值调低

Three
extrem
meanin
三重保
护。在

LARS Optimizer [Optimizer] — FAILED

将 LARS 学习率调低

LARS m
stagna
explos
LARS
173->1

Multi-Layer ALS (Batch + Sequential) [Algorithm] — FAILED

将 Multi-Layer ALS 学习率调低

Batch:
Instant
self-re
underc
批量：
顺序：
时 rank

A-PROBE (Low-Rank Bottleneck) [Architecture] — PARTIAL

将 A-PROBE 学习率调低

Insert
learns
bottlen
在 In
通过瓶

A-SYNC (Gradient Injection) [Gradient Injection] — SUCCESS

将 A-SYNC 学习率调低

ALS co
co-evo
Consta
conver
ALS 计
永远不
Protoco

LARS Optimizer: Reduces NaN but Does Not Converge

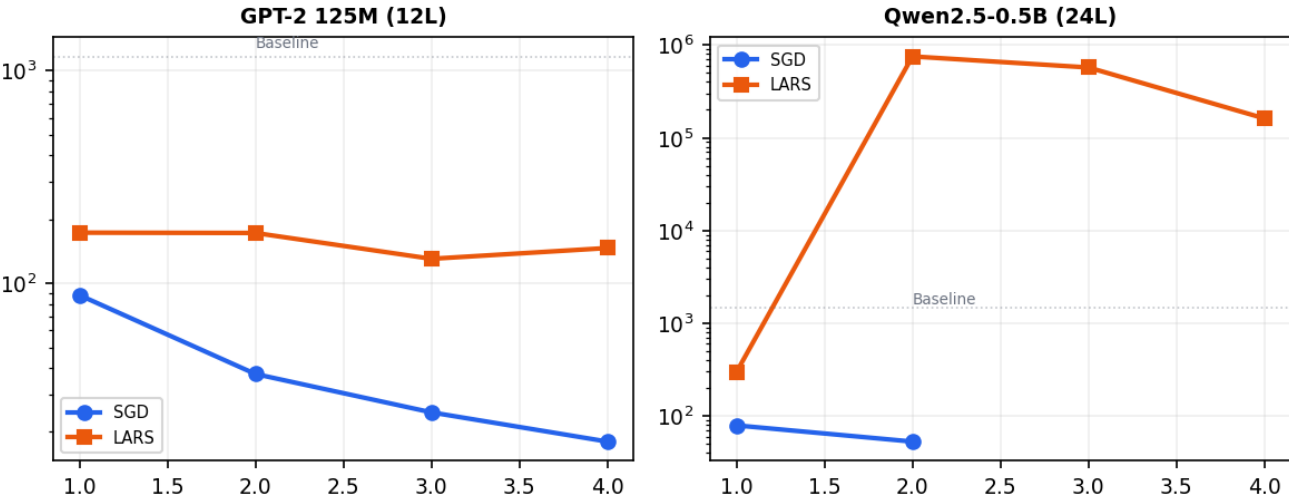


Figure 4: LARS vs SGD on GPT-2 12L (left) and Qwen0.5B 24L (right). LARS avoids NaN but does not converge.

4. A-SYNC Variant Progression

NDI 数据

Variant Evolution / :

1	A-SYNC +perturb	Gradient inject + noise	25.8	Converges, oscillates
2	A-SYNC no-perturb	Remove perturbation	16.6	Monotonic, cleaner
3	A-SYNC 16-cosine	Cosine sync decay	10.5	Good, sync dies early
4	A-SYNC 32-cosine	Cosine over 32 cycles	13.3	Decay kills tail
5	A-CYCLE restart	3x8 warm restart	16.5	Window too short
6	A-SYNC 24-const	Constant sync, 24 cycles	9.0	Excellent
7	A-SYNC 48-const	Constant sync, 48 cycles	7.6	BEST — converged

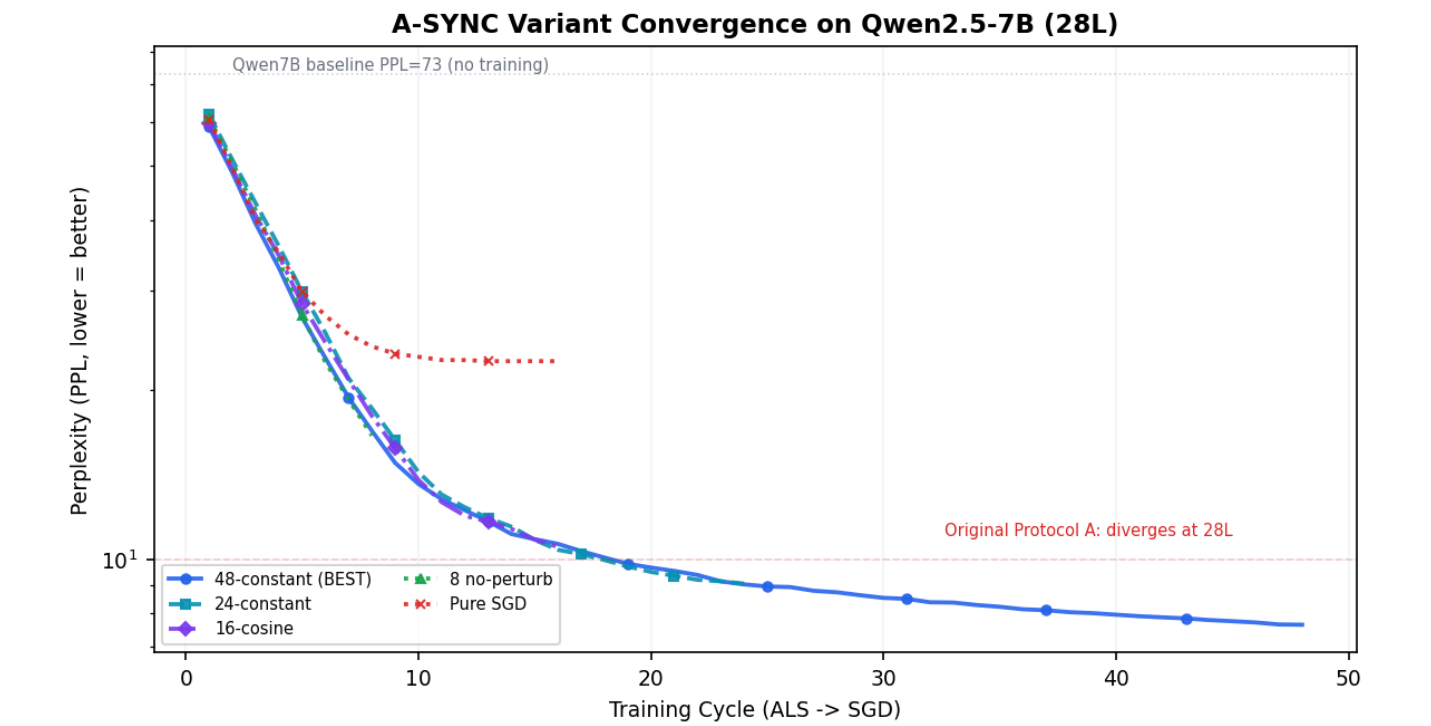


Figure 5: A-SYNC convergence curves on Qwen2.5-7B (28L). 48-constant sync (blue) is the clear winner.

Key discovery: constant sync strength (0.05) dominates all decay schedules. The ALS signal should persist at full strength —

5. Final Scoreboard

OD

All Approaches Compared / :

A-SYNC 48-constant (OURS)	Gradient Injection	7.6	CONVERGED
Pure SGD (no ALS)	Baseline	22.5	Plateaus
A-PROBE	Architecture	22.8	Bottleneck limited
A-KD	Distillation	195	KL explosion
LARS	Optimizer	161k	No convergence
Multi-layer ALS	Algorithm	DIVERGED	Underdetermined
Parameter Tuning	Tuning	DIVERGED	Non-transferable
Protocol A (original)	Baseline	DIVERGED	Residual amplification

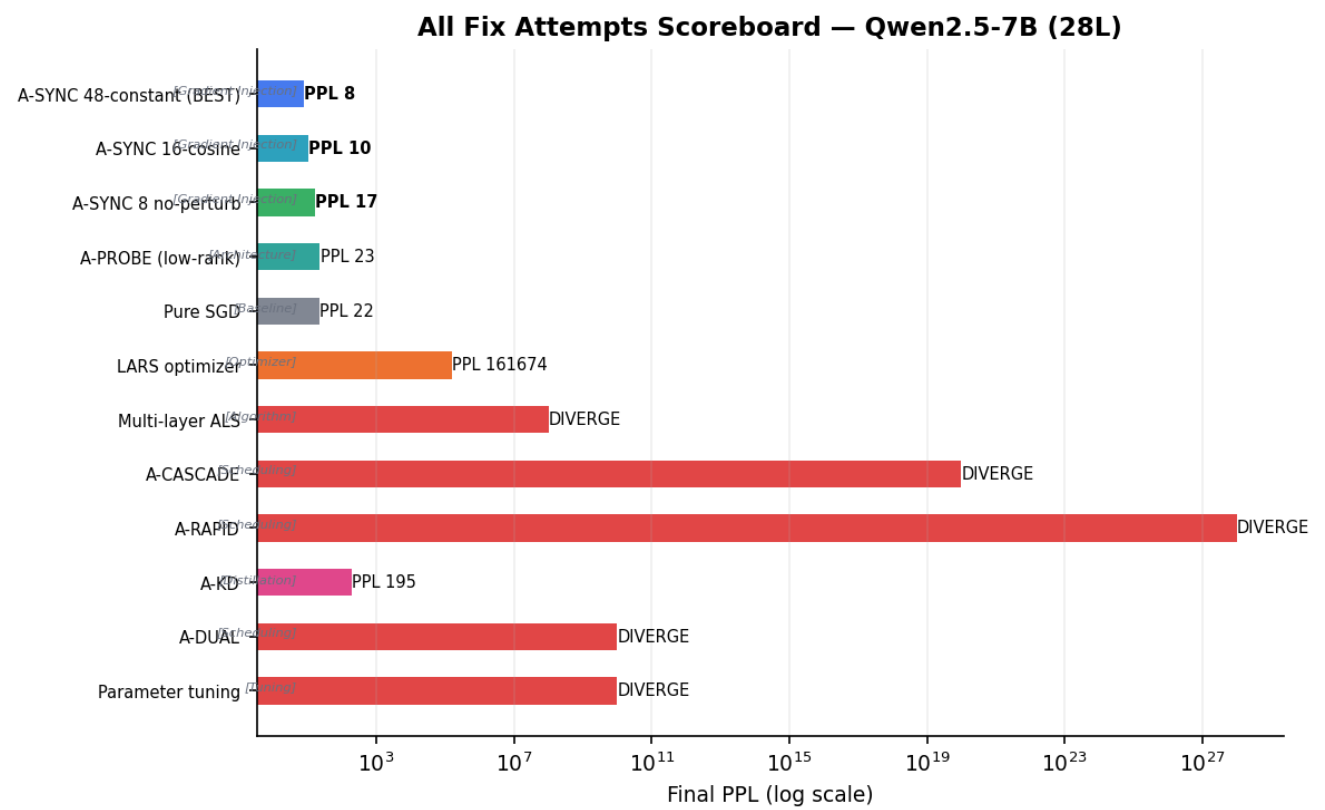


Figure 6: All fix attempts scoreboard — Qwen2.5-7B (28L). A-SYNC 48-constant is 3x better than pure SGD.

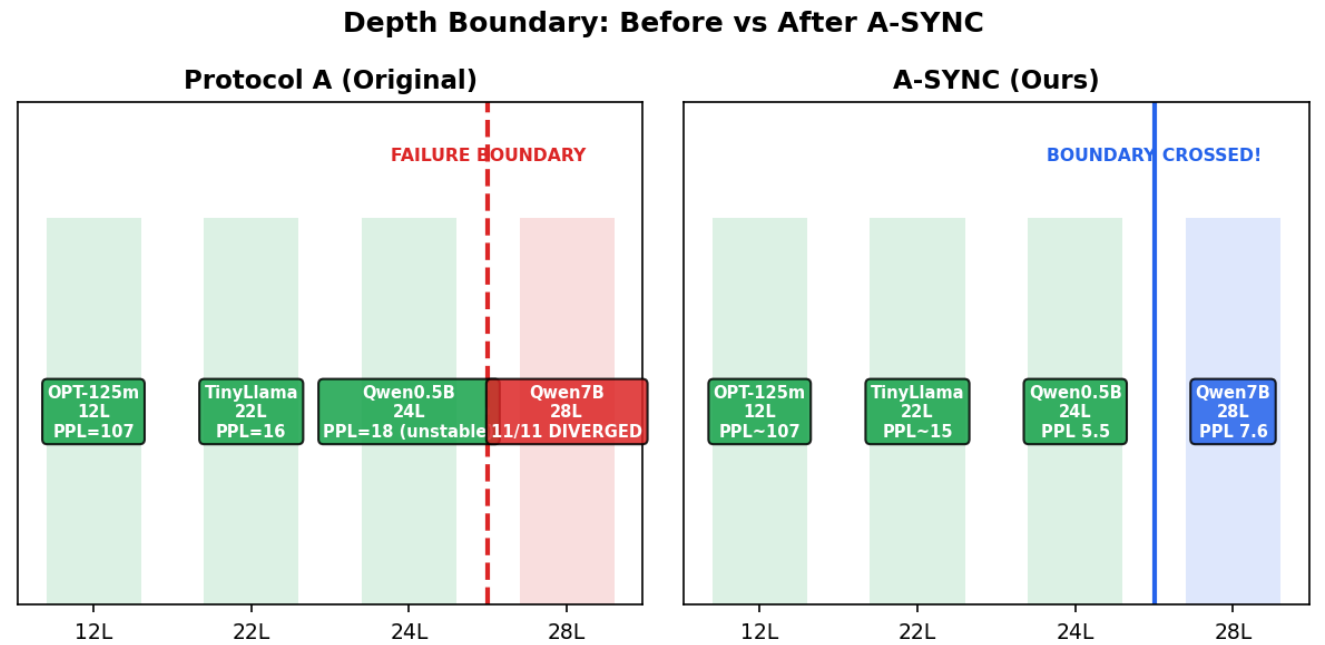


Figure 7: Protocol A (left) diverges at 28L. A-SYNC (right) crosses the boundary — monotonic convergence at all depths.